

PAWAN KUMAR DANGI

Imphal, Manipur, India

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Education

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, MANIPUR

B.Tech in Computer Science and Engineering (AIDS) — CGPA: 7.74

2023 – Present

Imphal, India

Experience

Centre for Development of Telematics (C-DOT)

May 19, 2026 – Jul 15, 2026

UAV Communication & Telemetry Intern

Onsite

- Worked on UAV communication and telemetry monitoring using **MAVLink**, studying swarm communication architectures, protocol-level security, and UAV-to-infrastructure communication.
- Developed a telemetry monitoring pipeline using **OMNeT++/C++** simulation, generating MAVLink telemetry and packet events and logging them into structured CSV files.
- Implemented a **Python rule-based anomaly detection engine** to monitor battery level, packet error rate, GPS fix, altitude bounds, and heartbeat timeouts and generate alerts.
- Built a **Streamlit dashboard** using Pandas to visualize live telemetry trends, UAV status, packet exchanges, and detected anomalies from telemetry and alert logs.

Projects

Adaptive Authentication System

Full-Stack Production Application

React, Vite, Tailwind CSS, FastAPI, PostgreSQL, Redis, SendGrid

[Live Demo](#)

- Built a production-grade authentication system with an intelligent risk engine that detects suspicious login patterns in real time and triggers multi-factor verification.
- Engineered risk scoring using behavioral signals including IP tracking, geolocation, device fingerprinting, temporal anomalies, and failed login attempts.
- Developed a FastAPI backend with JWT management, bcrypt password hashing, Redis-based rate limiting, PostgreSQL persistence, and SQLAlchemy ORM.
- Implemented React-based authentication flows, OTP verification, secure dashboard visualization, and automated deployment on Render.

Autonomous Intelligent Engineer (AIDE)

AI-Powered GitHub Issue Resolution Agent

FastAPI, Streamlit, RAG, Vector Embeddings, LLM Agents, Docker, GitHub Webhooks

[Github](#)

- Architected an autonomous AI agent that understands GitHub issues, analyzes repositories, identifies root causes, generates fixes, and prepares draft pull requests.
- Implemented **Retrieval-Augmented Generation (RAG)** using repository parsing and vector embeddings to retrieve contextually relevant code for issue resolution.
- Designed a multi-step agent workflow for issue reception, repository cloning, codebase analysis, root-cause identification, repair-plan generation, code modification, and validation.
- Integrated GitHub webhooks, automated repository operations, multi-LLM provider support, and Docker Compose for reproducible deployment.

Training & Certifications

- **AWS Cloud Practitioner – Training Badge** — Amazon Web Services (AWS), Jul 2026 ([Credential](#))
- **Business Intelligence & Analytics** — Indian Institute of Technology, Madras, Jan 2026 ([Credential](#))
- **Advanced Distributed Systems** — Indian Institute of Technology, Madras, Jul 2025 ([Credential](#))

Technical Skills

Languages: C++, Python, JavaScript, PHP, SQL

Frameworks & Libraries: Node.js, Express.js, NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn

Databases: MySQL, PostgreSQL

Tools & Platforms: Git, Linux, AWS, Vercel, Netlify, Render, Kaggle

Relevant Coursework: Operating Systems, DBMS, Computer Networks, OOPS, Compiler Design, Data Science, Machine Learning